WEEK 5 (Mon. / Thu.)

BEFORE YOU START — WHAT TO COLLECT

Inorganic Experiment 2, Session I — synthesis of crude ferrocene

THE ORDER DECIDES WHETHER THIS WORKS

Cyclopentadiene re-dimerises as it stands. **Collect the monomer last**, after the KOH/DME flask is assembled under N_2 *and* the $FeCl_2$ solution is already sitting in the dropping funnel. Assembling in any other order wastes the monomer. **Tell the lab officer, GTA or lecturer before two steps**: drawing the cyclopentadiene by syringe, and adding the concentrated HCl. Both are flagged in the manual.

A. Glassware and apparatus

☐ 100 mL **three-neck** (14/20) round-bottom flask
☐ 25 mL round-bottom flask — $FeCl_2$ solution
☐ Rubber septum
☐ Glass stopper
☐ $CaCl_2$ guard tubes × **2** + fresh $CaCl_2$
☐ Glass transfer gas adapter (N_2 inlet)
☐ Bubbler with nujol
☐ **Pressure-equalizing dropping funnel**
☐ Magnetic stirrer + stir bars × 2 + retriever
☐ Retort stand + **3** clamps and bosses
☐ Glass funnel — for the $FeCl_2$ transfer
☐ Syringe (2–5 mL) + **metal** needle, long
☐ 250 mL beaker — HCl and ice
☐ 25 mL beaker — reserved acid portion
☐ Large beaker or dish for the ice-water bath
☐ Glass rod
☐ **Büchner** funnel + filter paper of matching size
☐ Filter (suction) flask + vacuum trap + tubing
☐ 2 more clamps for the filtration set-up
☐ Measuring cylinders, 10 mL and 25 mL
☐ Sample vial + label — crude ferrocene for Week 6
☐ Spatula, weighing boat, watch glass

B. Reagents and consumables

☐ KOH pellets, **4.0 g**
☐ Anhydrous DME, **12 mL**
☐ $FeCl_2 \cdot 4\,H_2O$, **1.3 g**
☐ Anhydrous DMSO, **7 mL** + **3.5 mL** rinse
☐ Cyclopentadiene, **1.0 mL** — *collect last*
☐ Concentrated HCl, **10 mL**
☐ Ice, **30 g** plus more for the bath
☐ Deionized water, **10 mL** rinse + **3** × **20 mL** wash
☐ Acetone and methanol for washing glassware
☐ Sharps container for the used needle

C. Shared equipment — plan around the queue

☐ N_2 supply tap at your fumehood — check it works first
☐ Cyclopentadiene still (lab officers) — ask before approaching
☐ Vacuum line or pump
☐ **Stereomicroscope** at 14×, and the camera
☐ Analytical balance

D. Personal and paperwork

☐ Own lab coat and safety goggles
☐ Full-length trousers, covered shoes, high-cut socks
☐ Gloves; DMSO carries solutes *through* skin, so change them if splashed
☐ This bench sheet and the datasheet, printed
☐ Phone or NUS email, to receive the microscope image
☐ Calculator, permanent marker, S/N number

WHAT GOES WRONG, AND WHEN

Ice first, then acid. Pour concentrated HCl slowly onto 30 g of ice inside the fumehood; the reverse order boils and throws acid. **Most of the KOH will not dissolve** in 12 mL of DME, and that is expected — deprotonation happens at the solid–liquid interface, so stirring matters, not dissolution. Do not add solvent to "fix" it. **The $FeCl_2 \cdot 4\,H_2O$ must be fully dissolved** before transfer; salt left in the 25 mL flask is iron that never reaches the reaction. **Three clamps, not two** — flask, dropping funnel and bubbler all need holding, and a funnel full of DMSO solution is top-heavy.

요약. 순서가 결과를 좌우한다. 사이클로펜타디엔은 방치하면 다시 이량화되므로 KOH/DME 플라스크를 N_2 하에 조립하고 $FeCl_2$ 용액을 적하 깔때기에 채운 다음 마지막으로 받아 온다. 주사기로 단량체를 뽑을 때와 진한 HCl을 다룰 때는 담당자에게 먼저 알린다. 산은 얼음 위에 붓는다. KOH가 대부분 녹지 않는 것은 정상이므로 용매를 더 넣지 말고 교반에 신경 쓴다. $FeCl_2 \cdot 4\,H_2O$는 완전히 녹은 뒤에 옮긴다. 클램프는 플라스크, 적하 깔때기, 버블러까지 셋이 필요하다.

WEEK 5 (Mon. / Thu.)

INORGANIC EXPERIMENT 2 — SESSION I

Synthesis and Microscopic Image of Crude Ferrocene

Venue	General & Synthetic Chemistry Lab, Block S5 Level-1	**Instrument / technique**	Stereomicroscope, use of syringe
Time	10:00 a.m. – 4:00 p.m. (Mon. or Thu.)	**Source**	Manual v0724 + lecture notes

$$2\,C_5H_6 + 8\,KOH + FeCl_2\cdot 4\,H_2O \longrightarrow Fe(C_5H_5)_2 + 2\,KCl + 6\,KOH\cdot H_2O \qquad 2\,KC_5H_5 + FeCl_2 \longrightarrow Fe(C_5H_5)_2 + 2\,KCl$$

The manual prints the first equation with one KCl, which does not balance. Two are required.

READ BEFORE TOUCHING ANYTHING

The $FeCl_2$ solution must already be in the dropping funnel before cyclopentadiene goes into the three-neck flask. Prepare it first, or the monomer stands and re-dimerises. It is cracked by the lab officers in Session 0 and must be used immediately.

Degas the syringe and needle with N_2 in the receiver flask before drawing 1.0 mL: draw and expel ∼1.5–2 mL of N_2 three times, releasing outside the flask. Use a metal needle; it is sharp. **Inform the lab officer, GTA or lecturer before this step and before adding the concentrated HCl.**

Safety. KOH, conc. HCl, $FeCl_2\cdot 4\,H_2O$, DME and cyclopentadiene are toxic and corrosive; DME, cyclopentadiene and ferrocene are flammable; DMSO, $CaCl_2$ and ferrocene are harmful and irritating. **Add acid to ice, never the reverse**, inside the fumehood — fumes rise off the beaker. Aqueous metal-ion waste → designated container.

Procedure

(i) Potassium cyclopentadienide

1. 100 mL three-neck (14/20) RBF: rubber septum in one side neck, glass stopper in the other.
2. Weigh and add **4.0 g KOH pellets** through the middle neck *quickly*; cap with a $CaCl_2$ guard tube.
3. Add **12 mL anhydrous DME** from the dispenser; cap with the guard tube.
4. With stirring, swap the guard tube for the gas transfer adapter (N_2 + bubbler) and the glass stopper for the pressure-equalizing dropping funnel. Open the N_2 tap slowly; hold **∼1 bubble s^{-1} for 3–5 min**.
5. With stirring, add **1.0 mL cyclopentadiene** by degassed syringe through the septum. Stir **10 min**. *Most of the KOH stays undissolved — see Appendix F.*

(ii) Iron(II) chloride solution — prepare this first

6. **1.3 g $FeCl_2\cdot 4\,H_2O$** into a 25 mL RBF; cap with a $CaCl_2$ guard tube.
7. Add **7 mL anhydrous DMSO** with stirring; wait until *fully* dissolved.
8. Transfer to the dropping funnel through a glass funnel. Rinse with **∼3.5 mL anhydrous DMSO** and transfer.

(iii) Formation and work-up

9. Add the $FeCl_2$ solution **slowly, ∼20 drops min^{-1}**, into the stirred cyclopentadienide. Stir a further **30 min** under gentle N_2.
10. Meanwhile add **10 mL conc. HCl slowly to 30 g ice** in a 250 mL beaker. After 5 min transfer **10 mL** of it to a 25 mL beaker.
11. Stop the N_2; remove all glassware from the three-neck flask.
12. With a glass rod, *slowly* add the reaction mixture into the 250 mL beaker of HCl and ice, stirring.
13. Put the flask in an ice-water bath, add the reserved 10 mL with stirring, swirl, pour everything into the 250 mL beaker.
14. Rinse the flask with **10 mL DI water** and transfer. Stir **15–20 min**.
15. Collect on a **Büchner funnel**; wash with DI water (**3 × 20 mL**); dry under vacuum **∼30 min**. Record the mass.
16. Record a microscopic image at **14×** with your matriculation number typed on it. This is the "before" image; Week 6 supplies the "after".

Before you leave: ☐ Crude ferrocene, dried, in a labelled vial for Week 6 ☐ Microscopic image captured and sent to yourself ☐ Needle in the sharps container ☐ Datasheet signed by GTA

5주차 (월 / 목)

무기 실험 2 — 세션 I

조 페로센 합성과 현미경 이미지

장소	General & Synthetic Chemistry Lab, Block S5 Level-1	**장비**	실체현미경, 주사기
시간	오전 10시 – 오후 4시 (월 또는 목)	**출처**	매뉴얼 v0724 + 강의노트

시작하기 전에 반드시 읽을 것

$FeCl_2$ 용액이 적하 깔때기에 들어가 있어야 사이클로펜타디엔을 3구 플라스크에 넣을 수 있다. 순서를 바꾸면 단량체가 기다리는 동안 다시 이량화된다. 단량체는 세션 0에서 실험실 담당자가 분해해 둔 것이며 받는 즉시 써야 한다.
1.0 mL를 뽑기 전에 주사기와 바늘을 수용 플라스크에서 N_2로 치환한다. 1.5–2 mL를 들이마셨다가 플라스크 밖으로 내보내는 동작을 세 번 반복한다. 금속 바늘을 쓰고 날카로우므로 주의한다. **이 단계와 진한 HCl을 다루는 단계 전에는 실험실 담당자나 GTA, 강사에게 반드시 알린다.**
안전. KOH, 진한 HCl, $FeCl_2 \cdot 4\,H_2O$, DME, 사이클로펜타디엔은 독성·부식성이 있다. DME, 사이클로펜타디엔, 페로센은 인화성이며 DMSO, $CaCl_2$, 페로센은 유해·자극성이다. **산은 얼음 위에 붓는다. 반대로 하지 않는다.** 흄후드 안에서 하며, 제대로 해도 증기가 올라온다. 금속 이온이 든 수용액 폐기물은 지정 용기로 보낸다.

실험 절차

(i) 포타슘 사이클로펜타디엔화물

1. 100 mL 3구(14/20) 둥근바닥플라스크의 한쪽 옆목에 고무 셉텀을, 다른 옆목에 유리 마개를 끼운다.
2. KOH 펠릿 **4.0 g**을 칭량하여 가운데 목으로 신속히 넣고 $CaCl_2$ 건조관으로 막는다.
3. 무수 DME **12 mL**를 디스펜서에서 받아 넣고 건조관으로 막는다.
4. 교반하면서 건조관을 기체 전달 어댑터(N_2 + 버블러)로, 유리 마개를 압력 평형 적하 깔때기로 바꾼다. N_2 밸브를 천천히 열어 **초당 1방울 정도로 3–5분간** 유지한다.
5. 교반하면서 치환한 주사기로 사이클로펜타디엔 **1.0 mL**를 셉텀을 통해 가한다. **10분간** 교반한다. KOH는 대부분 녹지 않는다. 부록 F를 볼 것.

(ii) 철(II) 염화물 용액 — 이것을 먼저 준비한다

6. $FeCl_2 \cdot 4\,H_2O$ **1.3 g**을 25 mL 둥근바닥플라스크에 넣고 $CaCl_2$ 건조관으로 막는다.
7. 무수 DMSO **7 mL**를 교반하면서 가하고 완전히 녹을 때까지 기다린다.
8. 유리 깔때기로 적하 깔때기에 옮긴다. 무수 DMSO **약 3.5 mL**로 헹궈 함께 옮긴다.

(iii) 생성과 후처리

9. 교반 중인 사이클로펜타디엔화물에 $FeCl_2$ 용액을 **분당 약 20방울로 천천히** 가한다. 이후 N_2를 약하게 흘리면서 **30분간** 더 교반한다.
10. 그동안 250 mL 비커의 얼음 **30 g** 위에 진한 HCl **10 mL**를 천천히 붓는다. 5분 뒤 그중 **10 mL**를 25 mL 비커로 옮긴다.
11. N_2를 잠그고 3구 플라스크에 연결된 유리기구를 모두 뗀다.
12. 유리 막대로 저으면서 반응 혼합물을 HCl-얼음이 든 250 mL 비커에 천천히 옮긴다.
13. 플라스크를 얼음물 욕조에 담그고 남겨 둔 10 mL를 교반하면서 넣은 뒤, 흔들어 전량을 250 mL 비커에 붓는다.
14. 플라스크를 증류수 **10 mL**로 헹궈 옮기고 **15–20분간** 교반한다.
15. **Büchner 깔때기**로 모으고 증류수 **20 mL씩 3회** 세척한 뒤 진공에서 **약 30분간** 건조한다. 질량을 기록한다.
16. **14배**로 현미경 이미지를 찍고 학번을 이미지에 넣는다. 이것이 정제 전 이미지이며, 6주차에 정제 후와 비교한다.

퇴실 전 확인: ☐ 건조한 조 페로센을 라벨링한 바이알에 담아 6주차용으로 보관 ☐ 현미경 이미지 촬영 및 전송 ☐ 사용한 바늘 sharps 용기에 폐기 ☐ 데이터시트에 GTA 서명

DATASHEET — INORGANIC EXPERIMENT 2 — WEEK 5

Session I: Synthesis and Microscopic Image of Crude Ferrocene

Full name (as on student ID)	Matriculation no.	Date
Lab session ☐ Mon ☐ Thu	Fumehood no.	S/N no. assigned by lab officer

1. Reagents charged

Reagent	Nominal	Actual	Remarks
KOH pellets / anhydrous DME	4.0 g / 12 mL		
Cyclopentadiene monomer	1.0 mL		
$FeCl_2 \cdot 4\,H_2O$ / DMSO (dissolve, rinse)	1.3 g / 7, 3.5 mL		
Conc. HCl / ice	10 mL / 30 g		
DI water (rinse / wash 3×)	10 / 20 mL		

2. Conditions

	Nominal	Actual	Remarks
N_2 purge before CpH addition	3–5 min		
Stirring after CpH addition	10 min		
$FeCl_2$ addition rate / stirring after	20 drops min^{-1}, 30 min		
Stirring after acid quench	15–20 min		
Vacuum drying	∼30 min		

3. Colour checkpoints — two of these tell you the reaction is working

Stage	Expected	Observed / time	✓
KOH in DME, before CpH	white undissolved solid, colourless liquid		
10 min after CpH	**deep burgundy / red-brown.** A pale pink means deprotonation is incomplete		
$FeCl_2 \cdot 4\,H_2O$ in DMSO	**pale green.** Yellow or orange means Fe^{2+} is already oxidising		
During the $FeCl_2$ addition	darkens to a dark orange-brown slurry		
On quenching into ice-acid	orange precipitate; some gas evolution		
Filtrate and washings	pale yellow-green, fading to colourless		
Crude ferrocene, dried	orange to orange-brown powder		

The two bold rows are the checks worth making *before* committing the next reagent. A burgundy solution says the cyclopentadienide formed; a pale pink says stir longer. A green iron solution says the Fe^{2+} is still intact; if it has gone yellow or orange, oxidation has already cost you yield and the low result is explained. Write what you see, not what is printed.

4. Crude product

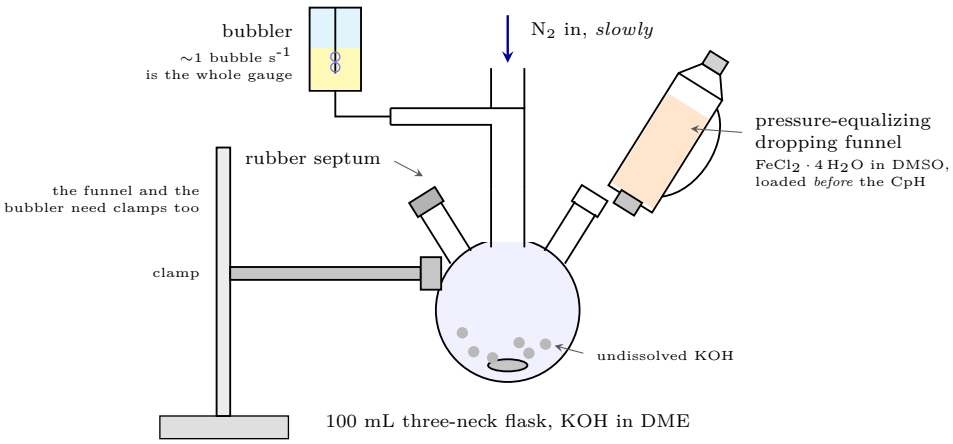
Mass of empty vial		N_2 flow maintained throughout ☐	
Mass of vial + crude		Gas evolution on quenching ☐	
Mass of crude ferrocene		Microscopic image taken ☐	

To be completed at the end of the session		
Student signature	GTA name	GTA signature

Raw data and observations must be signed by the GTA before leaving the laboratory. This datasheet is submitted as an appendix to the report.

Appendix F — Working Under Nitrogen

F1. The assembly



Why the funnel is pressure-equalizing. The side tube lets nitrogen above the flask reach the space above the liquid in the funnel. Without it, liquid draining out would pull a vacuum in the funnel and stop, or air would be sucked back in through the stopcock. **Why the bubbler.** It is the only indicator that gas is actually flowing and it caps the pressure inside the flask. About one bubble per second is enough; a faster stream just blows solvent out. **Why nitrogen at all.** Cyclopentadienide is air-sensitive, iron(II) oxidises readily to iron(III), and the cyclopentadiene monomer is volatile. All three fail in the same direction: less ferrocene.

F2. Drawing the monomer by syringe

1. Assemble the KOH flask under N_2 first and load the $FeCl_2$ solution into the dropping funnel. Only then approach the cyclopentadiene.
2. Inform the lab officer, GTA or lecturer before this step.
3. Push the needle through the septum of the receiver flask on the cyclopentadiene setup. **Degas: draw ~1.5–2 mL of N_2 into the syringe and expel it outside the flask. Repeat three times.** Air left in the barrel goes straight into the reaction.
4. Draw **1.0 mL** of monomer, withdraw, and add it through the septum of the three-neck flask with the stirrer running. Stir 10 min.
5. Rinse the syringe with acetone and put the needle in the **sharps container** — never the bin.

Use a metal needle. Plastic is only for acid work, and there is no acid at this stage.

F3. The chemistry the setup is protecting

Cracking dicyclopentadiene is a **retro-Diels–Alder** reaction, which is why the monomer reverts on standing: the forward cycloaddition is favourable at room temperature. Hence "use immediately".
KOH deprotonates cyclopentadiene at the sp^3 carbon. The anion that results is **aromatic** — a planar five-membered ring with six π electrons — and that aromatic stabilization is what makes an ordinary hydrocarbon acidic enough for hydroxide to deprotonate it at all. The same delocalization is what lets all five carbons bind iron in the η^5 mode.
The acid quench at the end has two jobs: it destroys unreacted cyclopentadienide and excess KOH, and it puts the product into water, in which ferrocene is insoluble. The ferrocene precipitates and everything ionic stays in solution — which is why the washing is done with deionized water.

요약. 적하 깔때기가 압력 평형형인 이유는 액체가 흘러내릴 때 깔때기 안이 진공이 되어 멈추거나 콕을 통해 공기가 빨려 들어가는 것을 막기 위해서다. 버블러는 기체가 실제로 흐르는지 보여주는 유일한 지표이자 플라스크 내부 압력의 상한 역할을 하며, 초당 한 방울이면 충분하다. 더 세게 흘리면 용매만 날아간다. N_2가 필요한 이유는 셋이다. 사이클로펜타디엔화 음이온은 공기에 민감하고, 철(II)는 쉽게 철(III)로 산화되며, 단량체는 휘발성이다. 세 경우 모두 결과는 페로센 감소로 같다. 주사기는 반드시 미리 N_2로 세 번 치환한다. 관 안에 남은 공기는 그대로 반응기로 들어간다. 마지막 산 처리는 미반응 음이온과 과량 KOH를 없애는 동시에 물에 녹지 않는 페로센을 석출시키는 두 가지 역할을 한다.